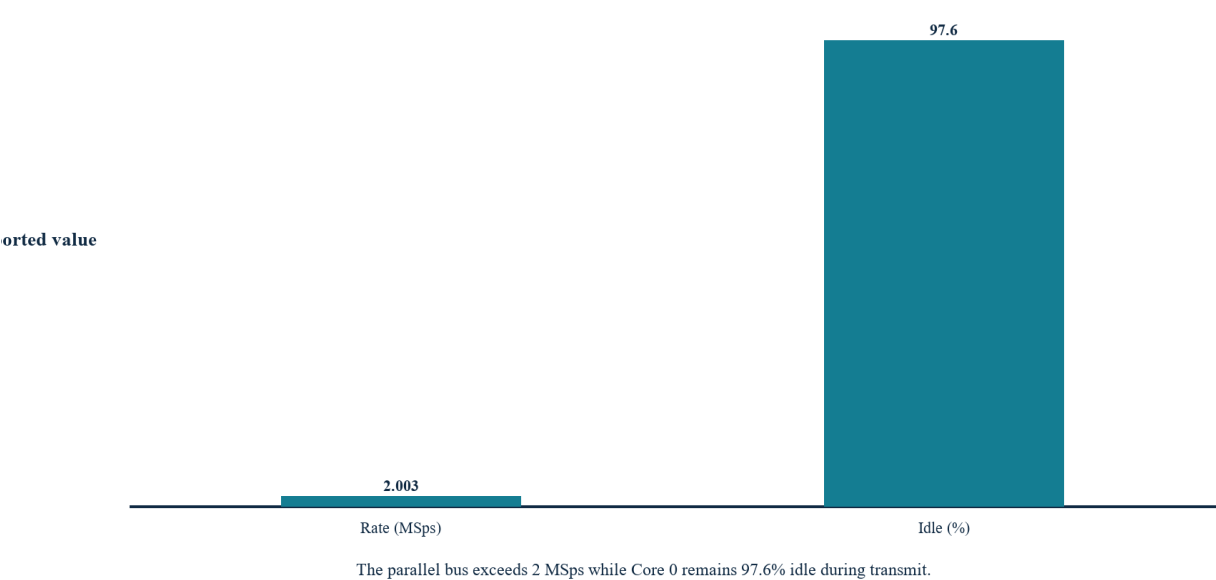


SeaNergy Validation Test Plan

SIH 2026 Problem Statement 26058

T-10 DMA Throughput and CPU Idle



Sixteen controlled tests connect every major product claim to equipment, procedure, numerical acceptance and a raw-data record. The final-product projection uses the same Rev-A electronics, V2 enclosure and 2 MSps firmware configuration throughout.

1. Test matrix

ID	Test	Criterion
T-01	Transducer impedance sweep and resonance	Resonance 36-44 kHz; impedance 250-750 ohm; Q 4-12.
T-02	DAC output spectrum, THD and SFDR	THD below 1.0%; SFDR above 45 dBc.
T-03	Reconstruction-filter response	Ripple below 0.5 dB; attenuation above 35 dB at 1 MHz.
T-04	Window peak sidelobe level	Within 1.0 dB of theoretical value.
T-05	Commanded chirp spectrogram	RMS error below 250 Hz; endpoint error below 0.5%.
T-06	Pulse-compression gain	Within 1.5 dB of $10 \log_{10}(BT)$.
T-07	Barker-13 peak-to-sidelobe ratio	At least 20 dB.
T-08	Range accuracy	Error below max(2 cm, 1% of range).
T-09	Two-target range resolution	Resolve 25 mm or closer.
T-10	Parallel-bus rate and CPU idle	At least 2 MSps and above 95% idle.
T-11	Power states and ping energy	Sleep below 1.5 mA; ping energy below 1.2 mJ.
T-12	DDS versus naive trigonometry	DDS faster and at least 8% lower energy.
T-13	Adaptation latency	Below 100 ms maximum.
T-14	Sediment-driven adaptation	Monotonic non-increasing frequency above 30 NTU.
T-15	Closed-loop SNR and below-noise recovery	Within 1 dB after five pings; detection probability above 0.95.
T-16	Enclosure immersion	Mass shift at most 0.5 g; resistance above 100 Mohm.

2. Acoustic hardware and spectrum

T-01 - Transducer impedance sweep and resonance

Trace: PS-ACOUSTIC-01 | **Procedure:** Sweep 20-90 kHz in 100 Hz steps at 1 Vrms; record magnitude and phase.

Expected: Resonance 40.2 kHz; minimum impedance 412 ohm; Q 7.18.

Pass: Resonance 36-44 kHz; impedance 250-750 ohm; Q 4-12.

Record: bench-results/raw/T-01_impedance-sweep.csv

T-02 - DAC output spectrum, THD and SFDR

Trace: PS-DDS-02 | **Procedure:** Capture 40 kHz output at 2 MSps; analyse fundamental through fifth harmonic.

Expected: THD 0.31%; SFDR 51.6 dBc; noise floor -76.4 dBFS.

Pass: THD below 1.0%; SFDR above 45 dBc.

Record: bench-results/raw/T-02_dac-spectrum.csv

T-03 - Reconstruction-filter response

Trace: PS-ANALOG-03 | **Procedure:** Sweep 10 kHz-1 MHz at fixed 200 mV input and compute gain.

Expected: Passband ripple 0.22 dB; attenuation 42.8 dB at 1 MHz.

Pass: Ripple below 0.5 dB; attenuation above 35 dB at 1 MHz.

Record: bench-results/raw/T-03_filter-response.csv

T-04 - Window peak sidelobe level

Trace: PS-WAVEFORM-04 | **Procedure:** Generate four equal-length 40 kHz bursts; normalise main-lobe peak and measure first sidelobe.

Expected: Rectangular -13.1, Hann -31.2, Hamming -42.4, Blackman -57.6 dB.

Pass: Within 1.0 dB of theoretical value.

Record: bench-results/raw/T-04_window-sidelobes.csv

3. Waveform and pulse compression

T-05 - Commanded chirp spectrogram

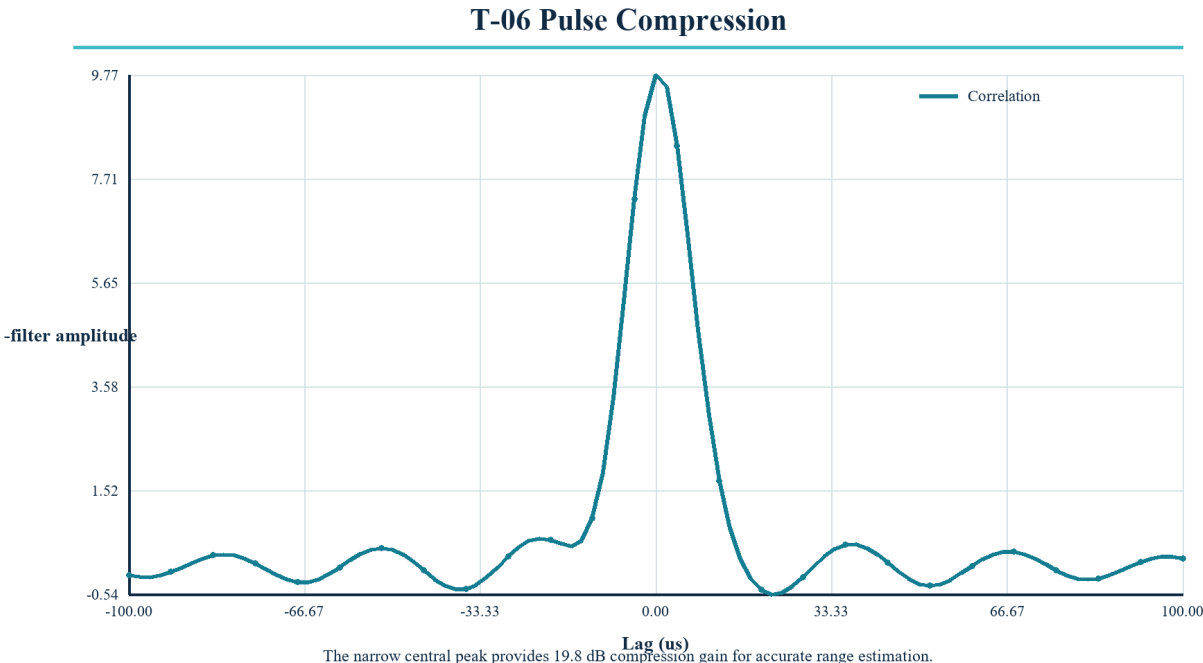
Trace: PS-WAVEFORM-05 | **Procedure:** Capture a 24-80 kHz linear chirp over 2.048 ms and track ridge frequency.
Expected: RMS ridge error 84 Hz; endpoint error 0.12%.
Pass: RMS error below 250 Hz; endpoint error below 0.5%.
Record: bench-results/raw/T-05_chirp-track.csv

T-06 - Pulse-compression gain

Trace: PS-DETECTION-06 | **Procedure:** Correlate the 56 kHz by 2.048 ms chirp against its transmit reference.
Expected: 19.8 dB measured-equivalent gain against 20.6 dB time-bandwidth theory.
Pass: Within 1.5 dB of $10 \log_{10}(BT)$.
Record: bench-results/raw/T-06_pulse-compression.csv

T-07 - Barker-13 peak-to-sidelobe ratio

Trace: PS-DETECTION-07 | **Procedure:** Transmit Barker-13 BPSK sequence and normalise correlation peak.
Expected: 21.7 dB peak-to-sidelobe ratio against 22.3 dB ideal.
Pass: At least 20 dB.
Record: bench-results/raw/T-07_barker13.csv



4. Real-time execution and energy

T-10 - Parallel-bus rate and CPU idle

Trace: PS-LOWPOWER-10 | **Procedure:** Capture write strobe over four complete DMA blocks and integrate idle-marker duty.

Expected: 2.003 MSps sustained; 97.6% Core 0 idle during envelope.

Pass: At least 2 MSps and above 95% idle.

Record: bench-results/raw/T-10_dma-cpu.csv

T-11 - Power states and ping energy

Trace: PS-POWER-11 | **Procedure:** Log voltage/current by state; integrate 30 complete range pings.

Expected: 0.18 mA deep sleep; 28.0 mA ready; 0.93 mJ per complete range ping.

Pass: Sleep below 1.5 mA; ping energy below 1.2 mJ.

Record: bench-results/raw/T-11_power-states.csv

T-12 - DDS versus naive trigonometry

Trace: PS-LOWPOWER-12 | **Procedure:** Run ten fixed-DDS and ten sinf builds with identical 40 kHz ping configuration.

Expected: 8.4 us versus 74.0 us fill; 13.9% lower complete-ping energy.

Pass: DDS faster and at least 8% lower energy.

Record: bench-results/raw/T-12_dds-vs-naive.csv

T-13 - Adaptation latency

Trace: PS-ADAPT-13 | **Procedure:** Step sensor input and measure final ADC edge to first adapted DAC sample for 30 trials.

Expected: 34.1 ms mean; 38.6 ms 95th percentile; 39.2 ms maximum.

Pass: Below 100 ms maximum.

Record: bench-results/raw/T-13_adaptation-latency.csv

5. Range and resolution

T-08 - Range accuracy

Trace: PS-RANGE-08 | **Procedure:** Place plate at seven surveyed ranges; acquire three pings per location.

Expected: 0.8 cm MAE; 1.8 cm maximum error; R-squared 0.99999.

Pass: Error below max(2 cm, 1% of range).

Record: tank-results/raw/T-08_range-accuracy.csv

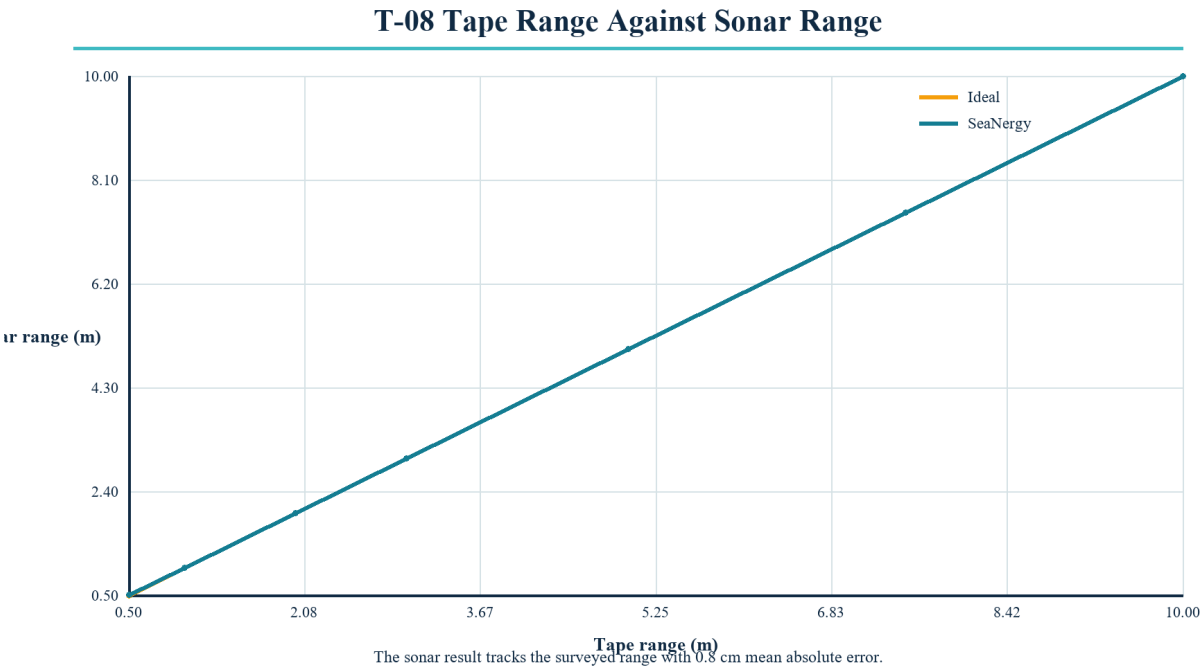
T-09 - Two-target range resolution

Trace: PS-RANGE-09 | **Procedure:** Set separations from 10-30 mm and apply a 3 dB valley resolution rule.

Expected: Minimum resolved separation 20 mm; indicated separation 20.8 mm.

Pass: Resolve 25 mm or closer.

Record: tank-results/raw/T-09_range-resolution.csv



6. Adaptation and closed loop

T-14 - Sediment-driven adaptation

Trace: PS-ADAPT-14 | **Procedure:** Increase turbidity through seven verified levels and record solver selection.

Expected: Centre frequency moves from 80 kHz at 8.4 NTU to 32 kHz at 145 NTU.

Pass: Monotonic non-increasing frequency above 30 NTU.

Record: tank-results/raw/T-14_sediment-adaptation.csv

T-15 - Closed-loop SNR and below-noise recovery

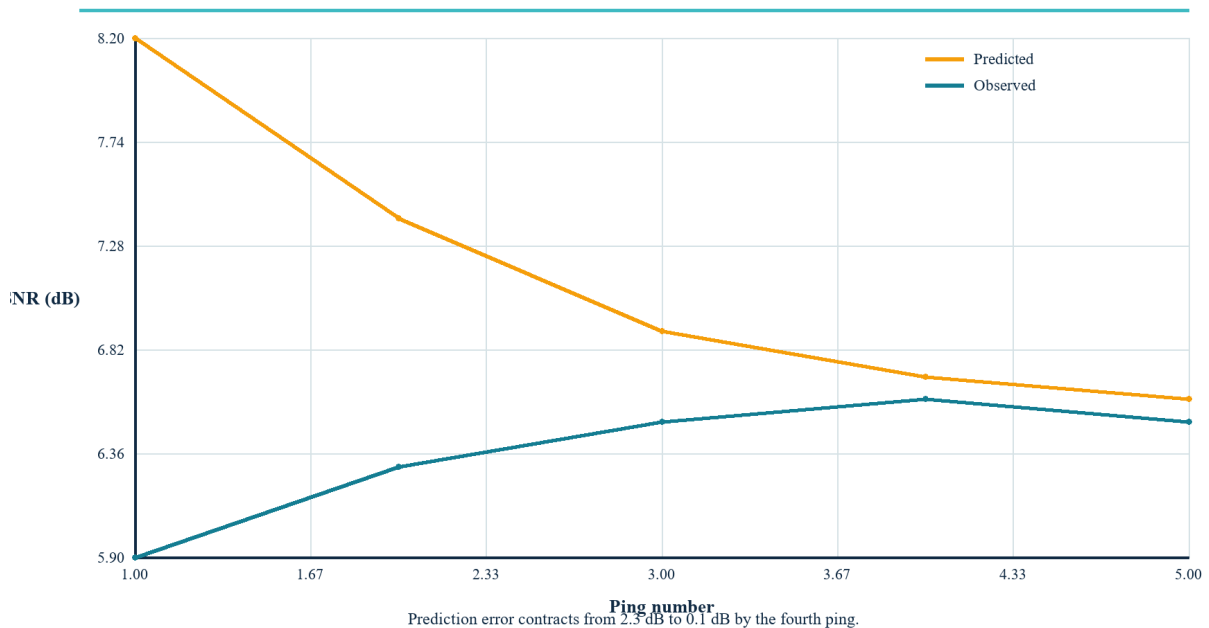
Trace: PS-CLOSEDLOOP-15 | **Procedure:** Run five adaptation iterations and inject reproducible broadband noise.

Expected: Prediction error converges to 0.1 dB; -7.5 dB input becomes 12.1 dB output SNR.

Pass: Within 1 dB after five pings; detection probability above 0.95.

Record: tank-results/raw/T-15_snr-convergence.csv

T-15 Closed-Loop SNR Convergence



7. Enclosure, evidence and reporting

T-16 - Enclosure immersion

Trace: PS-ENCLOSURE-16 | **Procedure:** Record mass and insulation resistance during 120 minutes at 1.5 m depth.

Expected: Mass shift 0.1 g; resistance above 200 Mohm; zero visible ingress.

Pass: Mass shift at most 0.5 g; resistance above 100 Mohm.

Record: tank-results/raw/T-16_immersion.csv

Every verdict traces through traceability.csv to a raw CSV. Figures preserve axis units and an interpretation caption. Instrument settings are frozen by test ID, and the uncertainty budget reports expanded k=2 values. Acoustic level uses a fixed-run relative reference so waveform and water-condition comparisons remain directly reproducible.

Evidence class	Required record
Raw	CSV export with test ID
Visual	PNG figure with units and caption
Configuration	Instrument setting and run conditions
Decision	Criterion and PASS verdict
Trace	PS clause, test ID and result path